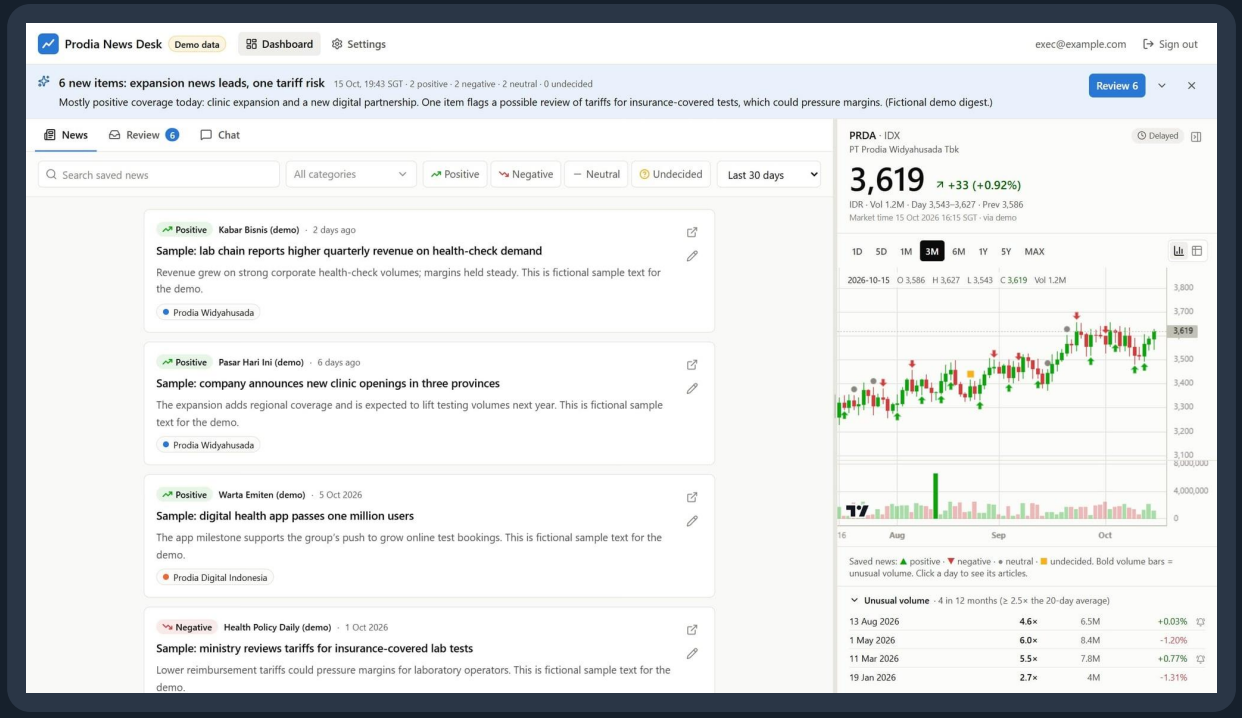
16 OCTOBER 2026

Prodia News Desk

Use case and overview of the implementation



Dashboard demo

OVERVIEW

What the News Desk does

Four jobs on one screen, plus a Telegram message when something needs you.

News

Finds news about Prodia every morning and tags each article: its topic and likely share-price effect.

Review

Nothing is saved until you approve it. Save, skip or correct the AI's tags in a click.

Chat

Ask questions in plain English. Answers cite your saved articles and PRDA prices.

Stock

PRDA price and volume, your saved news on the chart, and alerts for unusual trading.

Telegram: a digest after each run, plus alerts for unusual volume or failures.

UNDER THE HOOD

Frontend: the web app you see, running in your browser on a computer or phone.

Backend: services in Amazon Web Services (AWS) that search, run the AI and store data.

AI: Claude models by default; each task can switch provider.

Telegram bot: only sends messages with a link; it cannot change anything.

OVERVIEW

A day with the News Desk

The routine runs itself: you spend a few minutes reviewing, and ask questions whenever you need to.

All times are Singapore time (SGT).

06:00

The daily run searches the news published since the last run.

After the run

Telegram digest, e.g. “6 new items: expansion news leads, one tariff risk”.

Morning

You review the new items: save, skip or correct the AI.

10:00–17:45

PRDA price every 15 minutes, with an alert if volume spikes.

18:30

End-of-day price refresh and volume check.

Any time: ask the chat about the saved news and PRDA prices.

UNDER THE HOOD

EventBridge Scheduler: AWS’s alarm clock. Price refreshes follow IDX hours in Jakarta.

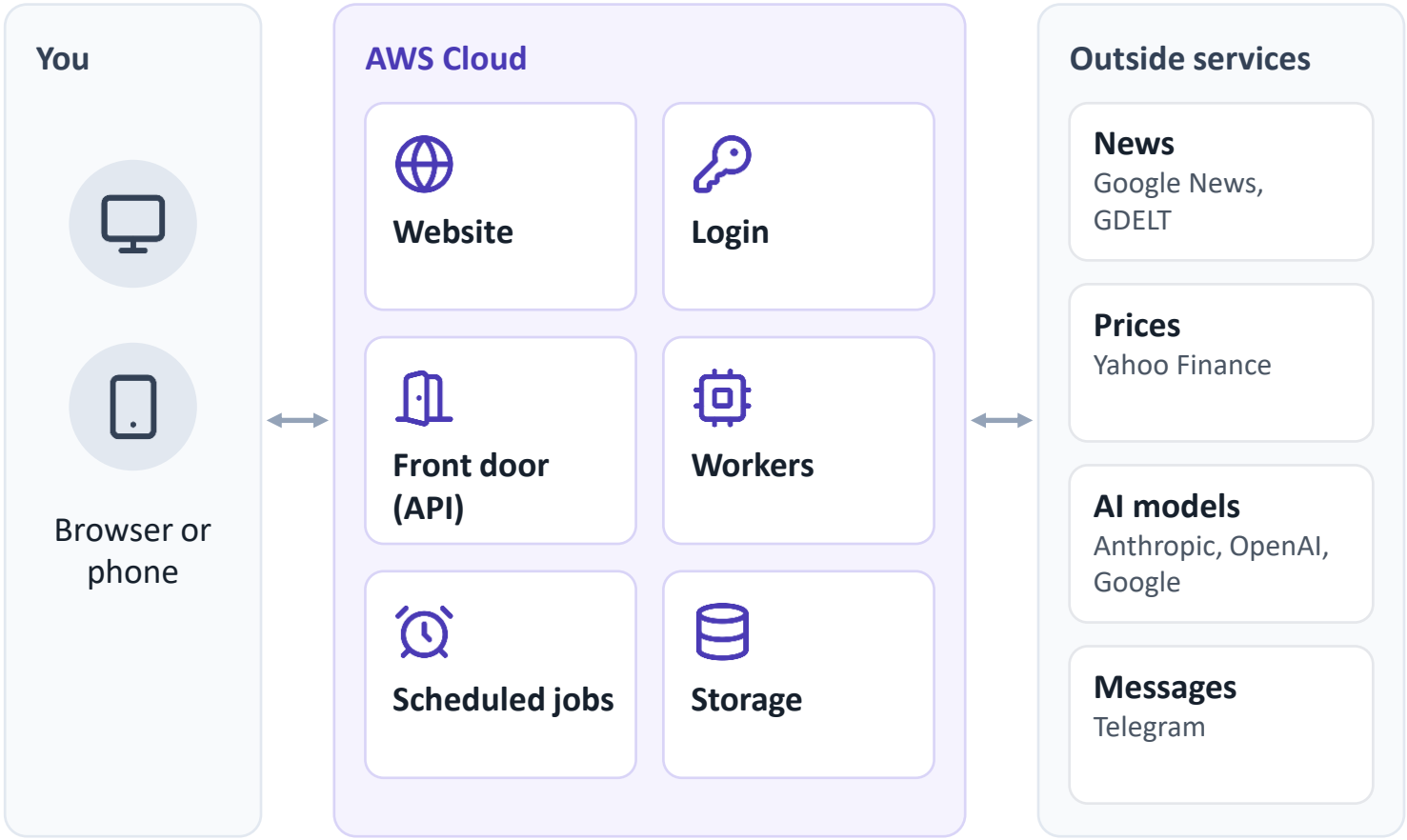
Editable: change the daily run time in Settings; the schedule updates itself.

Telegram: the digest has an “Open review queue” button that opens the Review tab.

OVERVIEW

The big picture

Your browser shows the screen, our part of AWS does the work, and outside services supply news, prices and AI.



UNDER THE HOOD

Website: CloudFront and S3 serve the app files.

Login: Amazon Cognito, invite-only.

Front door: API Gateway checks your login on every request.

Workers: AWS Lambda functions; Step Functions chains multi-step jobs.

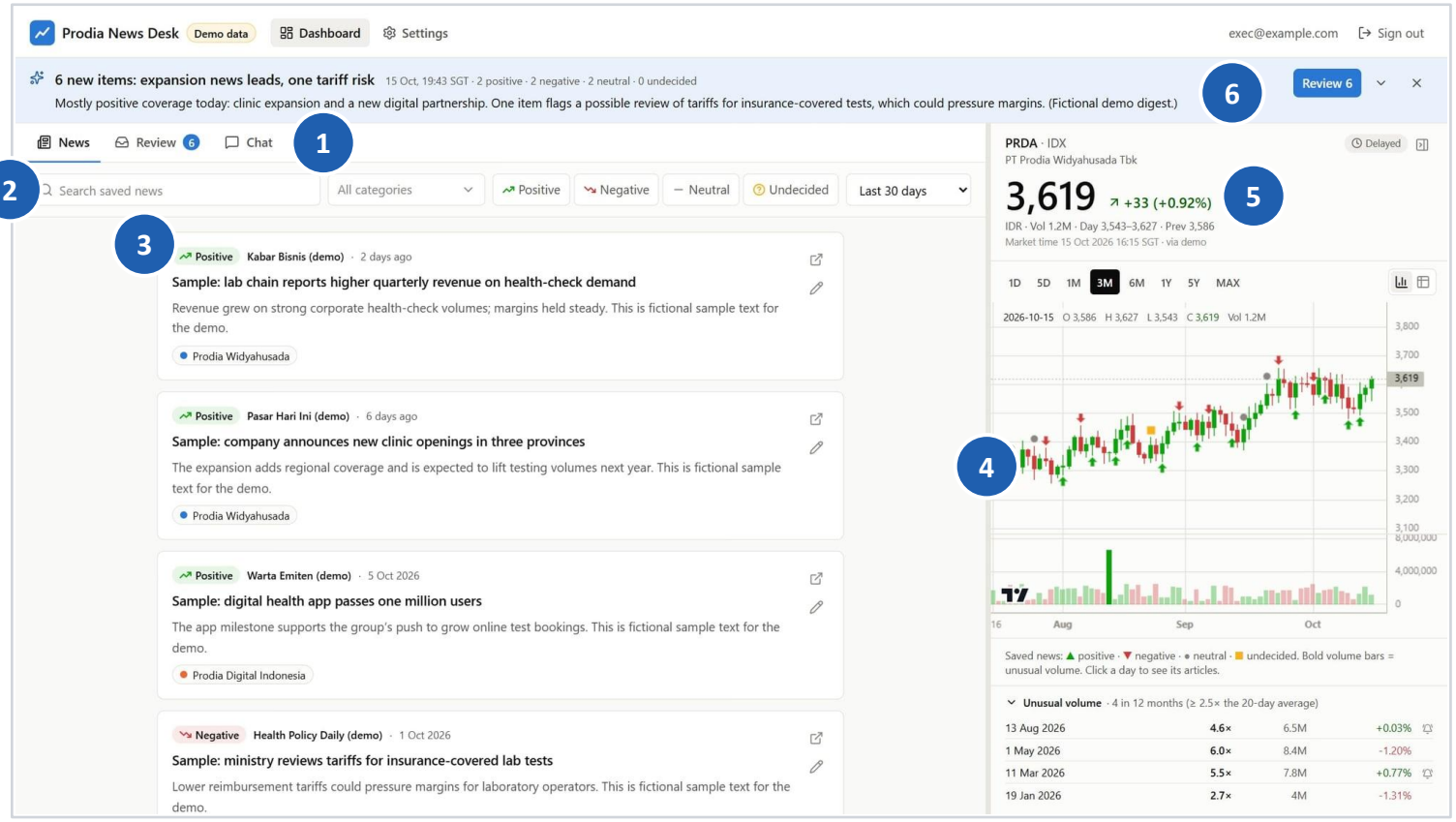
Scheduled jobs: EventBridge Scheduler.

Storage: DynamoDB, S3 and S3 Vectors; keys in Secrets Manager.

FRONTEND

The screen, piece by piece

One page: three tabs on the left, the PRDA panel on the right, and a divider you can drag between them.



Demo with fictional data

1. Tabs: News, Review, Chat
2. Search and filters
3. Article card with source link
4. Divider: drag to resize
5. PRDA price, chart and volume
6. The latest digest

UNDER THE HOOD

React: each part of the page is a reusable component; Tailwind styles it.

Layout: panel sizes are remembered; on narrow screens Stock becomes a 4th tab.

FRONTEND

How the screen gets its data

The app is downloaded once; after that, each tab fetches only small packets of data.

1

Open the link

CloudFront sends the app from a nearby server, so it loads fast.

2

Sign in

Cognito checks your password and gives the browser a digital pass.

3

Ask for data

Each tab asks the API, e.g. “saved news, last 30 days”, showing the pass.

4

Show and keep

The JSON answer is drawn on screen and kept, so tabs switch instantly.

What one saved article looks like as JSON

```
{ "title": "Ministry reviews tariffs for lab tests",
  "sentiment": "negative",
  "categories": ["Healthcare regulation"],
  "source": "Health Policy Daily" }
```

UNDER THE HOOD

Single-page app: one download, then only data travels (built with Vite).

TanStack Query: caches answers and refreshes them in the background.

Sign-in: OAuth 2.0 with PKCE; the pass is a signed token (JWT).

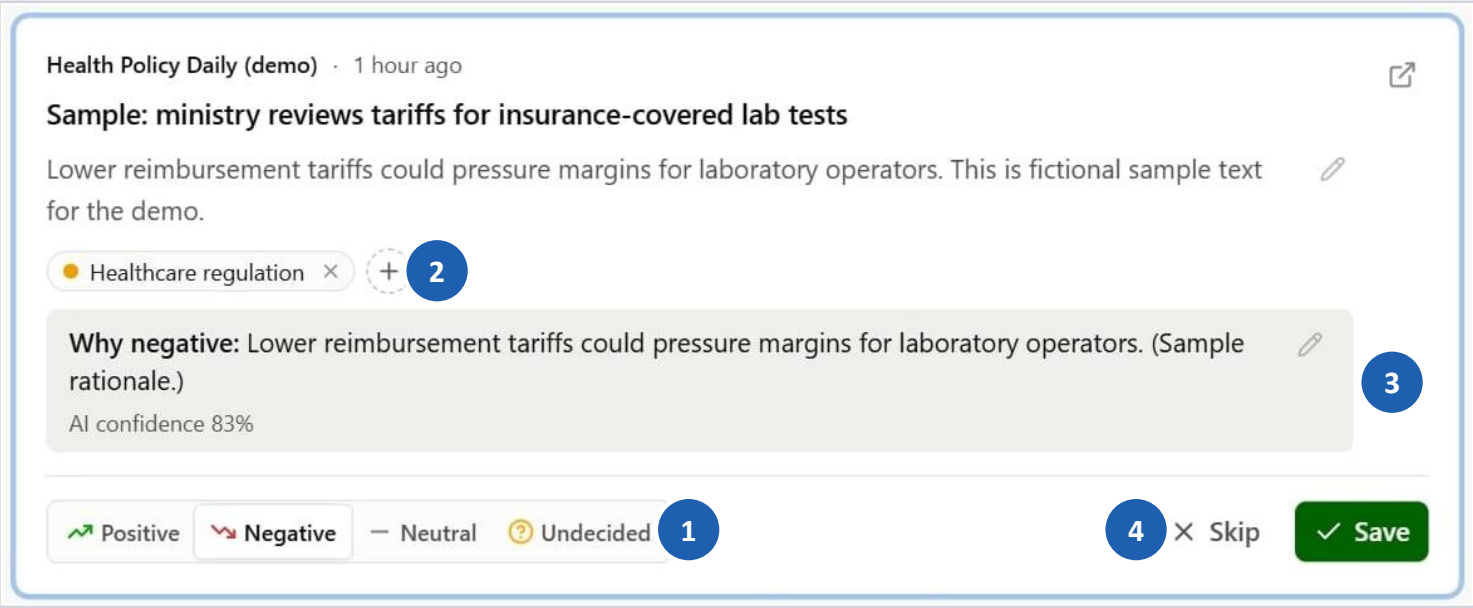
API Gateway: rejects any request without a valid pass.

config.json: lets one build run in test and production.

FRONTEND

Reviewing news

The AI proposes, you decide. Nothing reaches the archive without your approval.



1. Pick the sentiment
2. Remove (×) or add (+) topics
3. Edit the reason; see the AI's confidence
4. Save or Skip

A Save means it shows in News, the chat can find it, and it is kept as a training example.

UNDER THE HOOD

Shortcuts: J/K move, S saves, X skips, E opens details; "Save all as suggested" handles a batch.

Instant: each click sends PATCH /news/{id}; the screen updates at once and confirms in the background.

Feedback record: the AI's suggestion, your final answer and which fields you changed.

Memory: skipped stories are remembered, so they never come back.

BACKEND

Serverless: no servers to look after

AWS runs our code only when there is work to do and bills by the millisecond: under $10 a month at this scale.

Workers AWS Lambda	**Front door** API Gateway	**Conductor** Step Functions	**Alarm clock** EventBridge Scheduler	**Login** Amazon Cognito
Records Amazon DynamoDB	**Files** Amazon S3	**Delivery** Amazon CloudFront	**Key vault** Secrets Manager	**Alarms** Amazon CloudWatch

A traditional server is on all day, paid 24/7, and someone must patch it. **Serverless** wakes up for each task, scales by itself, and costs almost nothing when idle.

UNDER THE HOOD

Pay per use: billed by the millisecond, so quiet days cost almost nothing.

AWS CDK: the setup is code; one command rebuilds it in any AWS account.

cdk-nag: checks the setup against AWS security rules on every build.

BACKEND

The daily news run

Every morning a six-step workflow turns hundreds of search hits into a short list for you to review.

1	**Search** Google News and GDELT, in Indonesian and English	e.g. 300 hits
2	**Remove duplicates** The same article found twice, or seen or skipped before	120 new
3	**Quick AI screen** Headlines only; obvious noise is set aside for 30 days	30 relevant
4	**Read the article** Opens the publisher’s page and extracts the text	30 read
5	**AI tagging** Topics, sentiment, reason, summary and confidence	30 to review
6	**Digest** A short summary in the app and on Telegram	1 message

UNDER THE HOOD

Step Functions: runs the steps in order, with many searches in parallel.

Resilient: a failed search is skipped; a failed run is marked and you get a Telegram alert.

Backfill: when you add a topic, the same workflow searches past months, in windows of 7 to 30 days.

Sources: free Google News RSS feeds and the GDELT news database (last ~3 months).

BACKEND

How the AI tags an article

The AI gets clear instructions plus the article, and must answer in a fixed format that the app can check.

What the AI is given

- About Prodia and its group
- Your topic definitions
- The sentiment rules
- The article, marked as untrusted text



AI model

What it must return

Topic: Healthcare regulation

Sentiment: Negative

Why: lower tariffs could pressure margins

Summary: one or two sentences

Confidence: 83%

Positive

Likely to support the share price

Negative

Likely to pressure the share price

Neutral

Relevant, but unlikely to move it

Undecided

Could matter; the direction is unclear

UNDER THE HOOD

Structured output: the answer must match a fixed JSON format, checked (with zod) before it is saved.

Untrusted text: the article is fenced off as data, so instructions hidden in a web page are ignored.

Fallback: one retry, then a backup model if the answer still fails.

Usage: tokens are counted per run, for cost tracking.

Learning: optionally, your recent corrections are shown as examples (Settings).

BACKEND

Where the data lives

Four stores, each for one kind of data, all private and encrypted inside our AWS account.

DynamoDB · Records

Articles and tags, topics, runs, digests, prices, settings, your feedback.

S3 · Files

Full article text, chat conversations, raw search results (kept 90 days).

S3 Vectors · Meaning index

Saved articles as lists of numbers, so the chat can search by meaning.

Secrets Manager · Key vault

AI provider keys and the Telegram token, kept out of the code.

One trading day of PRDA, as stored in DynamoDB

PK (what)	SK (when)	open	high	low	close	volume
STOCK#PRDA.JK	D#2026-10-14	3,567	3,598	3,541	3,586	2.2M

Demo values. Slide 19 shows how rows like this become the chart.

UNDER THE HOOD

Single table: one DynamoDB table; each item's key says what it is (ART#…, STOCK#…, CATEGORY).

Auto-expiry (TTL): temporary items delete themselves, e.g. 15-minute price bars after 10 days.

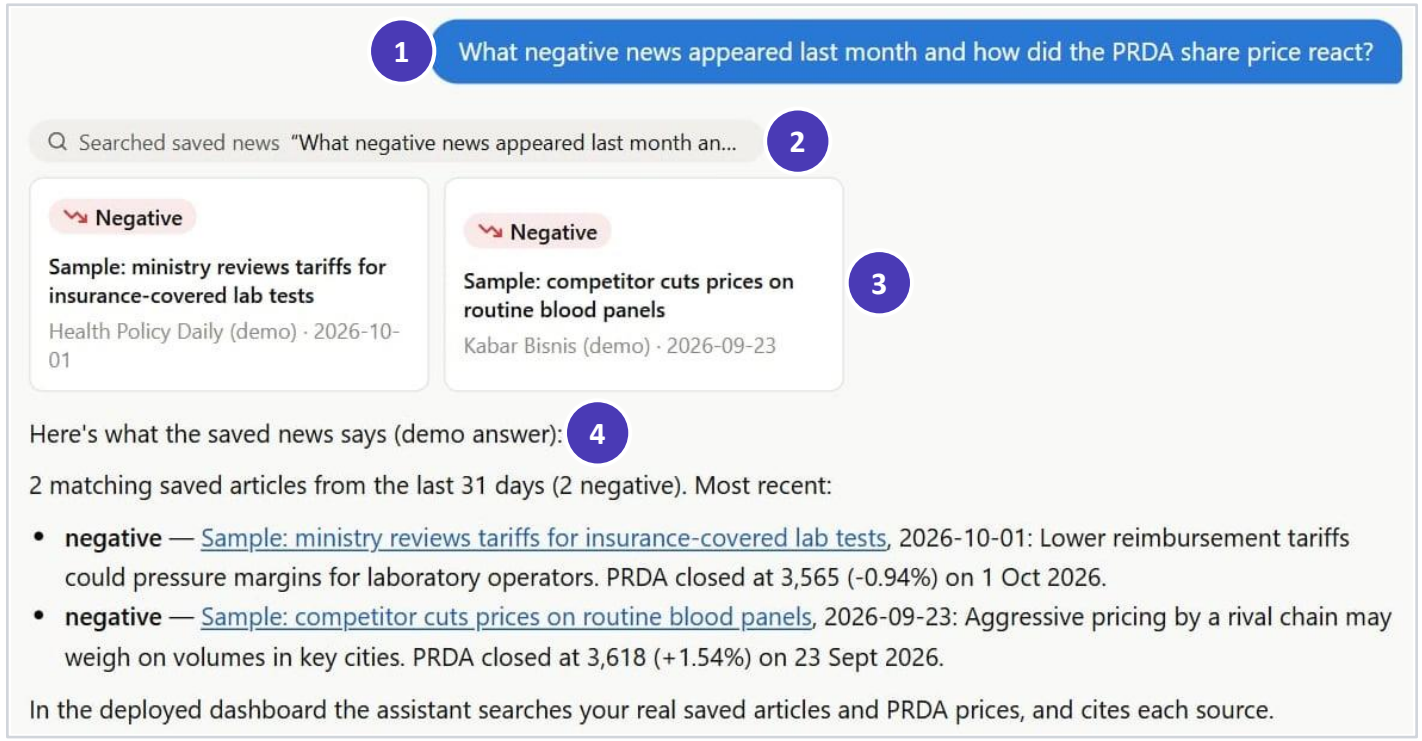
Private: encrypted storage that only the app's own functions can read.

Auto re-index: saving or editing an article updates the meaning index (DynamoDB stream → indexer).

BACKEND

The chat assistant

It looks things up before it answers: it searches your saved news, checks prices, then replies with sources.



1. Your question
2. It searches the saved news
3. The articles it found
4. An answer that cites them

UNDER THE HOOD

Search by meaning: each article becomes 768 numbers that capture its meaning, so "tariff cut" also finds "lower BPJS reimbursement" (Gemini Embedding 2 in S3 Vectors).

Tool calling: the model asks for searchNews or getStockPrices; our code runs them and returns the results.

Read-only: the assistant cannot change or delete anything.

Streaming: words appear as they are written; conversations are saved per user.

BACKEND

Stock prices and unusual volume

Prices refresh every 15 minutes during trading hours, and you get an alert when trading volume jumps.



Stock panel (demo data)

When prices update (Mon–Fri, Singapore time)



Every 15 minutes during the session (about 15 minutes delayed), then one end-of-day refresh at 18:30.

Unusual volume

2.5× the 20-day average volume, or more, triggers:

- a bold bar on the chart
- a line in the Unusual volume list
- one Telegram alert that day

UNDER THE HOOD

Source: Yahoo Finance through yahoo-finance2: free but unofficial.

Stored: one DynamoDB record per trading day, plus 15-minute bars kept for 10 days.

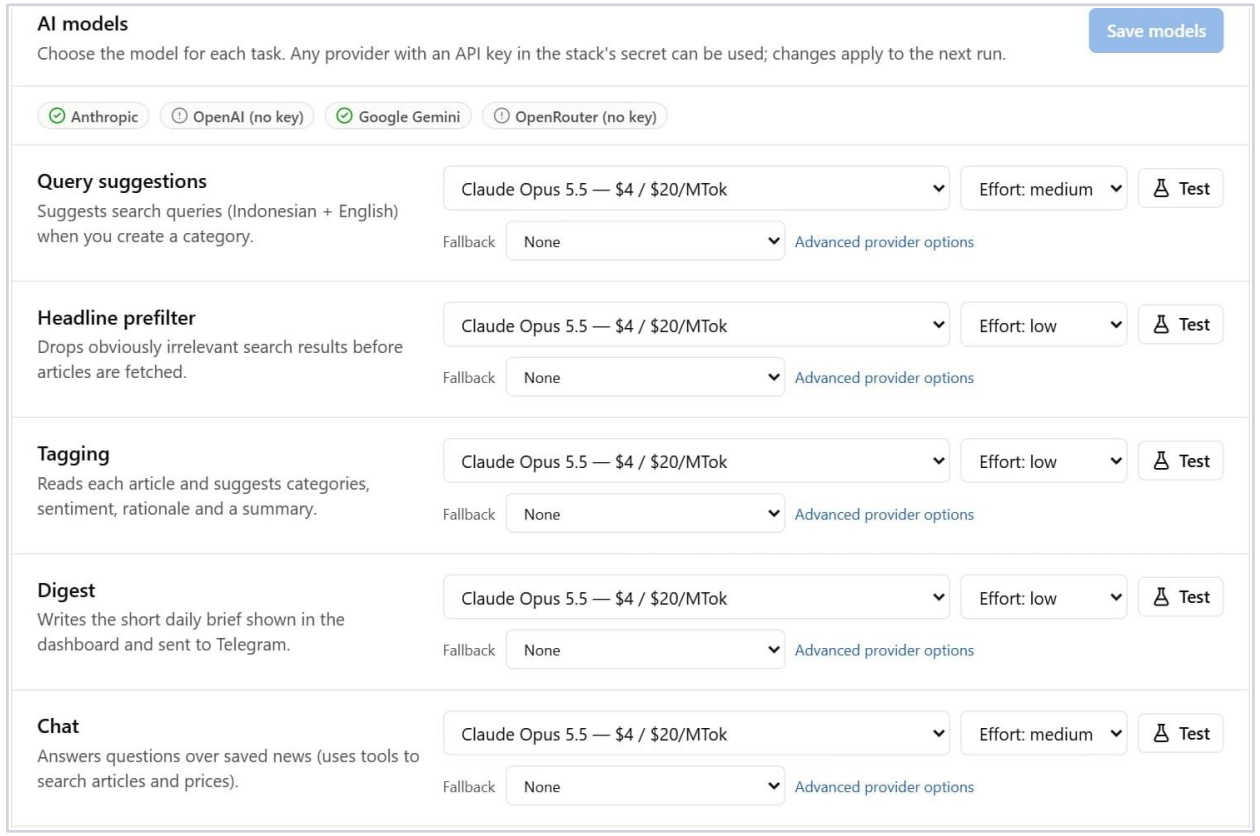
Safety net: after 3 failed refreshes, a “Not updating” badge and a Telegram alert.

Upgrade: one setting switches to EODHD, a paid feed ($19.99/month).

BACKEND

Choosing AI models

Not locked to one AI company: each task can use a different model, chosen in Settings.



Settings → AI models (demo). A green tick means that provider's key is set.

UNDER THE HOOD

Vercel AI SDK: one programming interface for Anthropic, OpenAI, Google and OpenRouter (hundreds of models with one key).

Per task: a main model, a backup model, and how hard it should think ("effort").

Before switching: the Test button checks a choice; eval:tagging scores a model against your past decisions.

Cost lever: tagging costs about $30–45 a month with Claude Opus, and about $1–3 with Claude Haiku.

Custom model id: any model a configured provider offers can be typed in, such as a brand-new release.

OVERVIEW

Security and privacy

Only invited people get in, every request is checked, and the AI can read your data but never change it.

Invite-only

No public sign-up; optional authenticator-app code.

Every request checked

Your login pass is verified on every API call.

Private and encrypted

Storage is private; data is encrypted in transit and at rest.

Keys in a vault

AI and Telegram keys live in Secrets Manager, not in code.

The AI only reads

Chat tools are read-only; article text is treated as untrusted.

Safe web fetching

The article reader refuses internal network addresses.

UNDER THE HOOD

Cognito authorizer: on every API route, verified by an automated test.

Least privilege: each function can reach only the data it needs.

cdk-nag: AWS security rules checked on every build.

Origin Access Control: the website files can only be fetched through CloudFront.

OVERVIEW

What it costs and what could break

Running costs are small; the main risk is that the free data sources are unofficial, and each one has a paid fallback.

Monthly running cost

< $10 — AWS: hosting, database and workflows

$0 — News and price data, and Telegram

$30–45 — AI tagging with Claude Opus (Haiku: $1–3)

$0.10 — Roughly, per chat question

What could break, and the fallback

Risk	What happens then	Paid fallback
Google News links are unofficial and can break	We keep the Google link; GDELT gives direct links	Serper: $50 per 50,000 searches
Yahoo Finance may block cloud servers	“Not updating” badge and a Telegram alert	EODHD: $19.99/month
GDELT only covers about 3 months	Older backfills rely on Google News	NewsData.io archive: from $199.99/month

Under the hood: token usage is recorded per run (Settings → Daily run). Prices as researched in October 2026.

OVERVIEW

Where we are and what's next

Built and tested, with a live demo on fictional data; going live on AWS is the next step.

Done

- Dashboard, news pipeline, chat and stock panel are built
- 96 automated tests and the AWS security checks pass
- Live demo with fictional data: news-desk-demo.vercel.app

Next

1. Deploy to AWS Singapore
2. Add the AI and Telegram keys
3. Invite you as a user
4. Connect Telegram
5. Add the topics to follow

Later ideas

- Let other AI tools query the archive (MCP)
- Save or skip straight from Telegram
- Group near-duplicate stories
- Watch IDX company disclosures

>_ Going live is one command (npm run deploy) once the review is done.

YOUR QUESTIONS

How the chart turns numbers into a picture

A CSV is just text. Excel shows it as a grid; our chart library draws it as candles.

1 As text: the CSV file

```
date,open,high,low,close,volume
2026-10-09,3552,3591,3483,3520,1400000
2026-10-12,3520,3554,3499,3517,537500
2026-10-13,3517,3605,3494,3567,1200000
2026-10-14,3567,3598,3541,3586,2200000
```

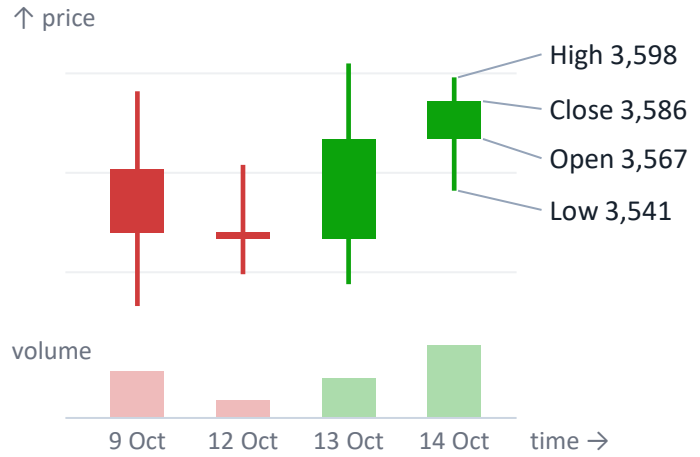
What the file really contains: plain text. No columns, colors or chart; each program decides how to show it.

2 As a grid: what Excel shows

Date	Open	High	Low	Close	Vol.
9 Oct	3,552	3,591	3,483	3,520	1.4M
12 Oct	3,520	3,554	3,499	3,517	538K
13 Oct	3,517	3,605	3,494	3,567	1.2M
14 Oct	3,567	3,598	3,541	3,586	2.2M

The same text shown as rows and columns. In the app, the stock panel's table button shows exactly this view.

3 As a drawing: our chart



Body = open to close; wick = high to low. Green: it closed higher than it opened. Red: lower.

UNDER THE HOOD

Stored: one DynamoDB record per trading day; the chart stores nothing.

Drawn: by TradingView's open-source Lightweight Charts, on an HTML canvas.

Redrawn: on every zoom or hover; date → left-right, price → height.

YOUR QUESTIONS

Open-weight vs closed-weight models

Weights are the billions of numbers a model learned in training. The difference is whether you can download them.

A model is a giant file of numbers (“weights”), set during training. Using the model means doing arithmetic with them.

70 billion numbers ≈ 140 GB file

Closed-weight

e.g. Claude, GPT, Gemini

- Used as a service: send text, get an answer back (API)
- The company keeps the weights and runs the hardware
- Pay per use; nothing to maintain
- Usually the most capable models

Open-weight

e.g. Llama, Qwen, DeepSeek, Mistral, Gemma, gpt-oss

- Download the weights and run them on your own servers or cloud
- Can be fine-tuned on your own data, which stays in-house
- You pay for GPUs and upkeep instead
- Each model has its own license

UNDER THE HOOD

Open-weight ≠ open source: usually only the weights are published, not the training data or code. A few are fully open, e.g. AI2’s OLMo.

In News Desk: defaults are closed Claude models; any task can switch to an open-weight model through OpenRouter, e.g. openrouter:deepseek/deepseek-chat.

Self-hosting: possible on AWS (SageMaker or Bedrock); it swaps per-use fees for GPU costs and upkeep.

The math: at 16-bit precision each number takes 2 bytes: 70 billion × 2 bytes ≈ 140 GB.

YOUR QUESTIONS

Building our own model with open-source tools

Nobody starts from scratch: you take an open-weight model and teach it your task with your own examples, which the app already collects.

Four options; effort and cost rise to the right →

1 Examples in the prompt

Today: the AI sees your recent corrections. No training.

Cost: ~$0 extra

2 Small classifier

Turn articles into numbers and train a simple model on your decisions.

Cost: low

3 Fine-tune an open model

Teach a small Llama or Qwen with LoRA on a rented GPU for a few hours, then host it.

Hosting a GPU: hundreds of $/month

4 Train from scratch

Trillions of words and millions of dollars. Not sensible for us.

Cost: millions

How a project would run



UNDER THE HOOD

Tools: Hugging Face Transformers, PEFT (LoRA) and TRL; Unsloth or Axolotl to fine-tune faster; scikit-learn for a classic classifier.

Our data: every Save, Skip and edit is stored with the AI's suggestion; npm run export:feedback exports it.

Yardstick: eval:tagging scores any model, including a home-grown one, against your past decisions.

When: once a few hundred to a few thousand articles are reviewed, and a cheaper model with examples is not accurate enough.